Thalvego

The operational 3D model of operating sites

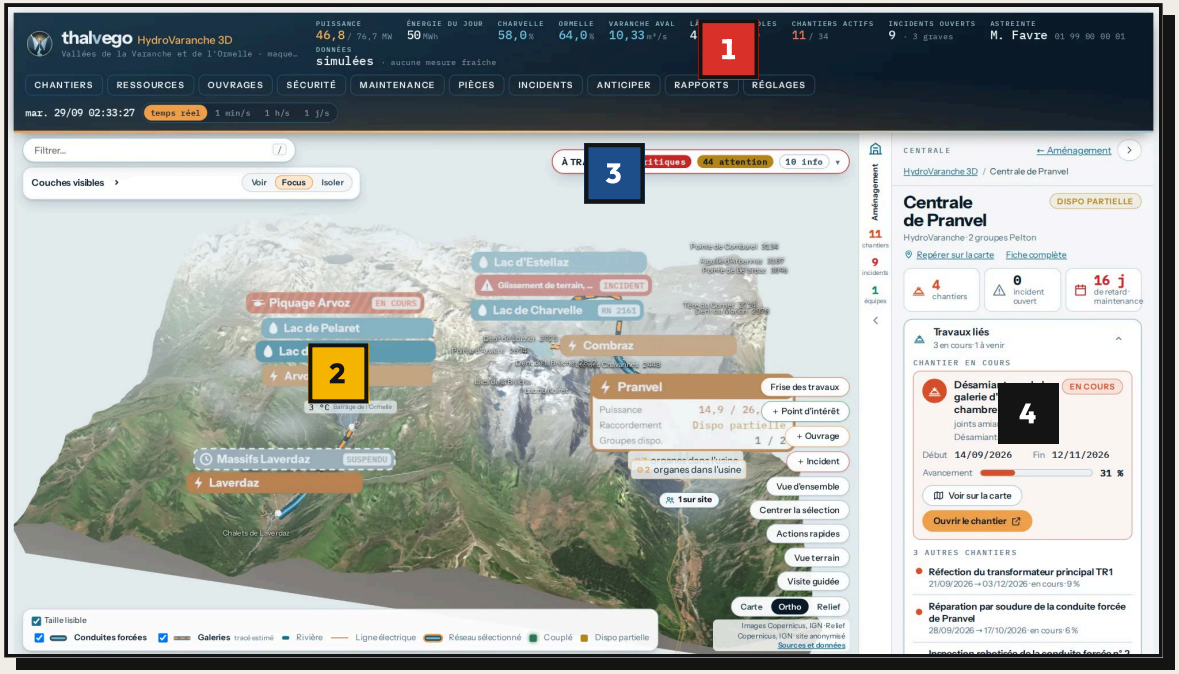
OPERATIONS · HYDROPOWER

1 The site at a glance

Power, energy, reservoir levels, active worksites, open incidents and on-call staff, at the top of the screen.

2 The valley on its terrain

Structures, worksites, incidents and reservoirs placed where the work happens. Layers are shown on demand.



Overview, on an anonymised site with simulated data.

3 To handle

Alerts ranked critical, warning, info: co-activity, lockouts, overdue worksites.

4 Structure and worksite card

Condition, deadlines, progress of related works and maintenance backlog, with direct access to the worksite.

WHAT THE TOOL COVERS

Assets
Structures, characteristics, periodic inspections.

Works
Worksites, progress, documents, reports.

Safety
Co-activity, lockouts, access and rescue.

Resources
Teams, on-call duty, authorisations, workload plan.

Weather
Temperature, rain and wind at each structure.

Risks and generation
Incident register, generation simulated for the least impact.

WHAT I DESIGNED

One shared engine, replicated site by site, with optional modules: weather, works and safety, resources, risks, simulated and measured generation. The showcase website and its film.

TOOLS

Web application, 3D on real terrain, editable reference data, Netlify website.

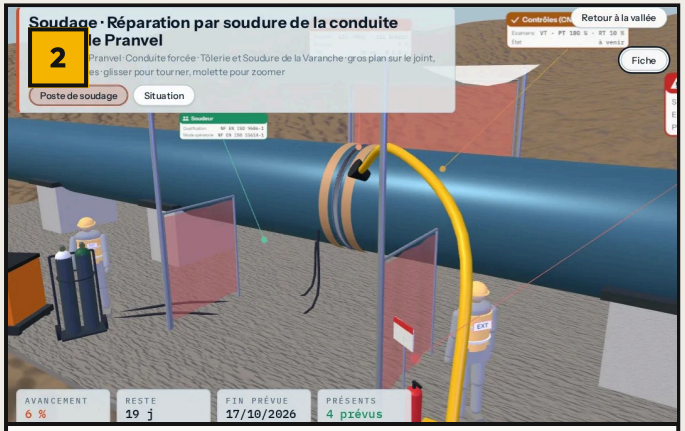
Thalvego

Each worksite opens in detailed 3D



The valley on its real terrain

Lakes, dams, power plants and penstocks laid on IGN terrain and aerial imagery. Worksites and incidents appear as labels; one click opens the worksite in detailed 3D.



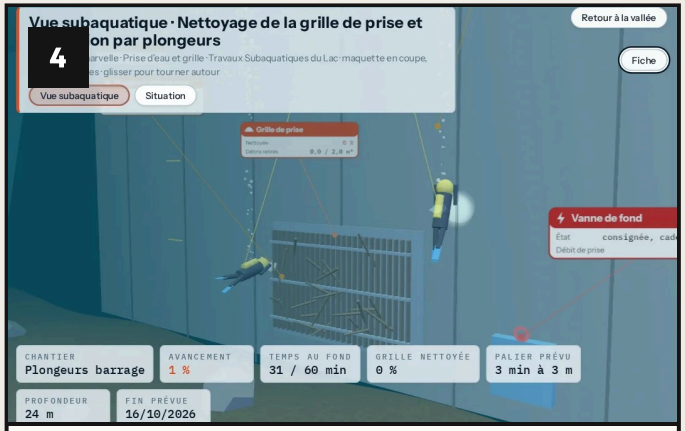
Welding a penstock

Welding station, qualified welder, planned inspections, marked-out area, worksite progress.



Overhauling a generating unit

Turbine and generator in the power house, crew on site or expected, impact on generation.



Divers on a water intake

Bottom time, depth, planned decompression stop and share of the screen already cleaned.

WHAT I DESIGNED

The 3D engine and its library of 135 models (dams, power plants, penstocks, cranes, helicopter, rope access, crews); one detailed scene per worksite, linked to its card: footprint, installations, crew, safety, progress.

TOOLS

Three.js, IGN and Copernicus terrain and aerial imagery, simulated demonstration data.

QME References · 03

BikePulse Gear

Quote and invoice built on the bike, in 3D

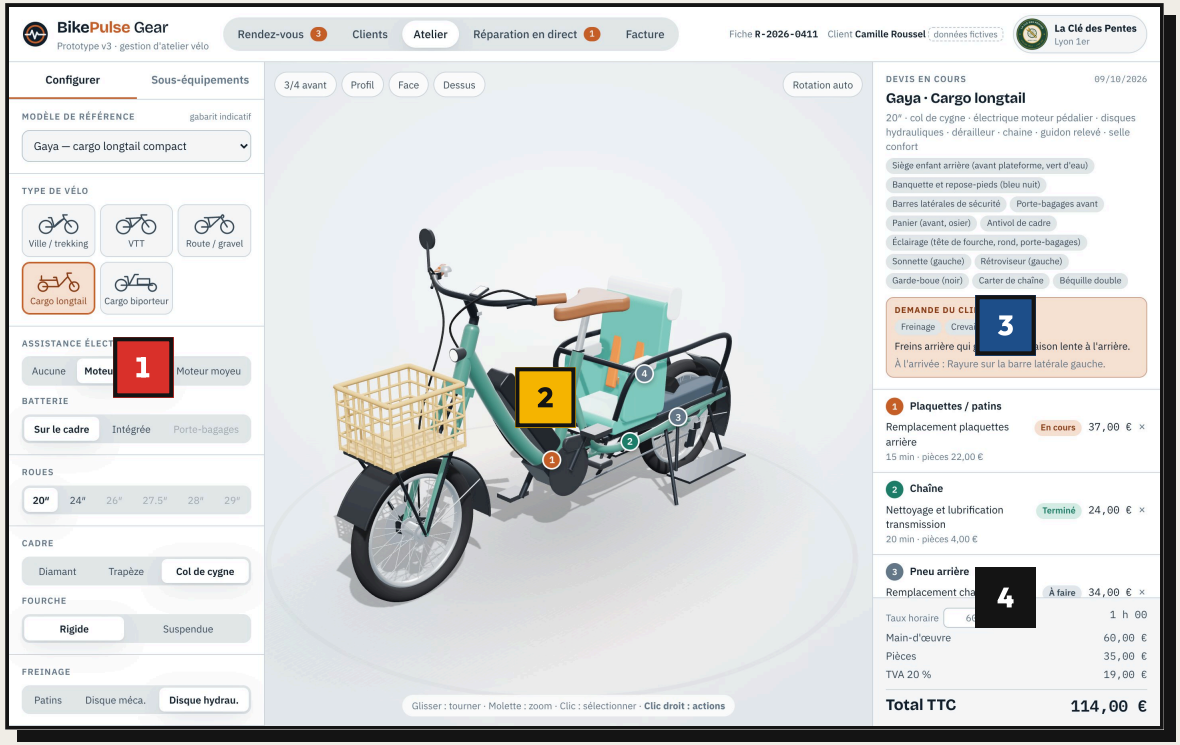
BIKE SHOPS · E-BIKES · CARGO

1 The customer's bike

Type, motor, battery, wheels, frame, fork, brakes: each choice rebuilds the 3D model. The reference model serves as a template.

2 Click on the part

Click on the bike, right-click for the action wheel: the job is added and a numbered marker pins it to the part.



Workshop screen. Fictitious workshop, customer and prices.

3 Customer request

Reasons for the visit and condition on arrival, entered at booking, stay in front of the mechanic.

4 A live quote

Time, parts and status for each job (to do, in progress, done). Labor, VAT and total are recalculated live; the invoice follows from it.

WHAT I DESIGNED

The full workshop flow: booking, bike drop-off, workshop, live repair, invoice. The 3D engine for the bike and its equipment. The presentation website with a narrated video and a form.

TOOLS

Single-page web app, Three.js for 3D, Netlify website, video edited by script with a synthetic voice-over.

Bricabulle

Nursery management with accurate billing

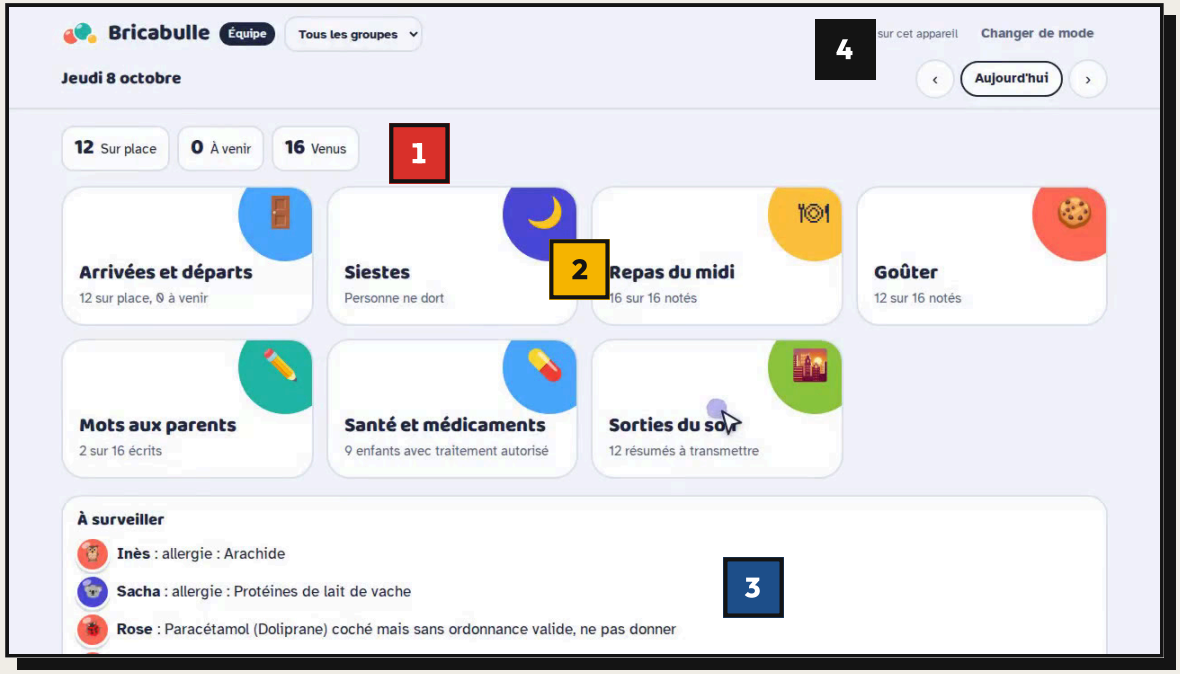
NURSERIES · EARLY CHILDHOOD

1 The room at a glance

Children present, expected and already gone: the team always knows who is there.

2 One tile per action

Arrivals and departures, naps, meals, snacks, notes to parents: each moment of the day is logged in one tap on the room tablet.



Team tablet. Fictitious children and nursery.

3 Health first

Allergies and treatments surface on their own. A medicine ticked without a valid prescription is flagged before it is given.

4 Four modes, one tool

Team tablet, parent kiosk at the entrance, online parent portal, management dashboard.

WHAT I DESIGNED

The link between the schedule booked by parents, the hours actually attended and the invoice. The evening summary for each child. Management oversight: real-time staff ratios, missed check-ins, disputes, schedules and leave.

TOOLS

Single-page web app, script-generated demo, subtitled video, website and form on Netlify.

Nexora

Tasks, projects and meetings, with AI agents plugged in

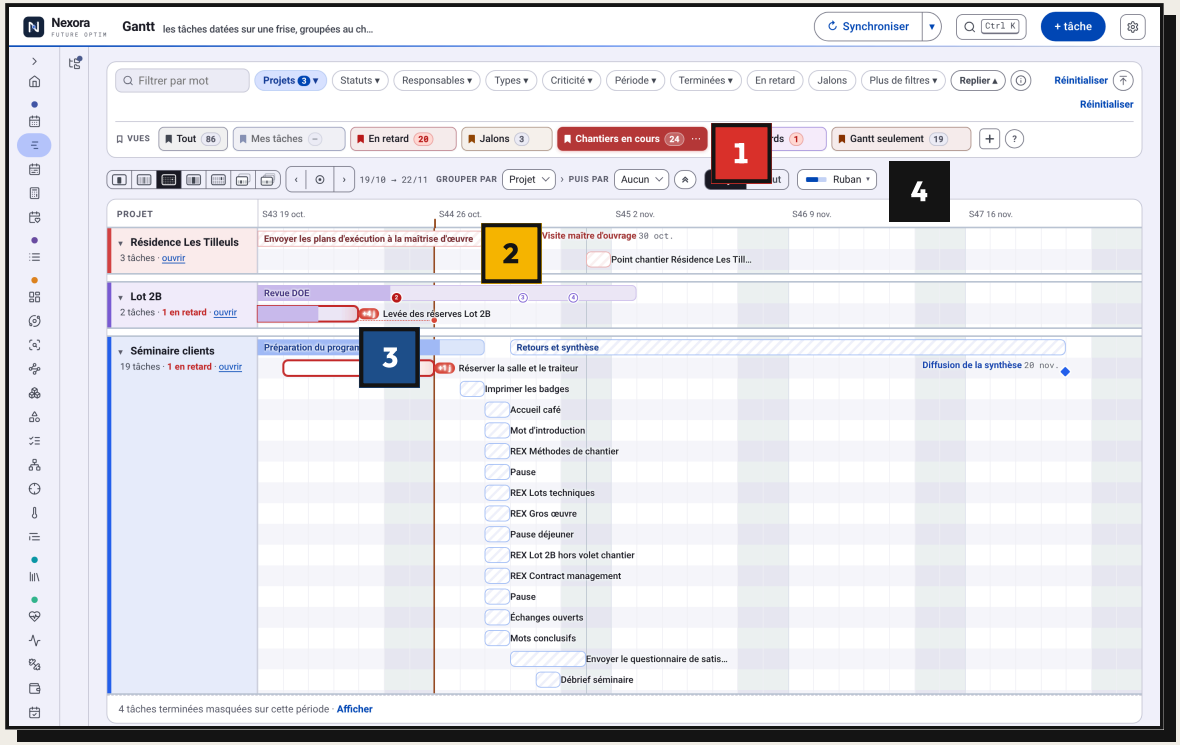
PROJECT MANAGEMENT

1 Saved views

A view keeps its filters. One click shows ongoing sites, another shows delays or milestones.

2 Milestones and visits

Visits, meetings and deadlines sit on the timeline, within their project. The vertical line marks today.



Gantt view of the app. Fictitious projects, tasks and people.

3 Visible delays

An overdue task is circled in red, with its delay in days. The bar shows progress.

4 Group, zoom

By project or by status, from week to year. The same data is also available as a list, an orbit or a daily dial.

WHAT I DESIGNED

A complete app: configurable task types, overlap-free scheduling, meetings and minutes, Google Calendar sync, a connector that lets AI assistants create tasks from incoming emails.

TOOLS

React, Firebase, Netlify, MCP connector, automated tests in a real browser.

Drawing approvals

From drawing to approval for construction

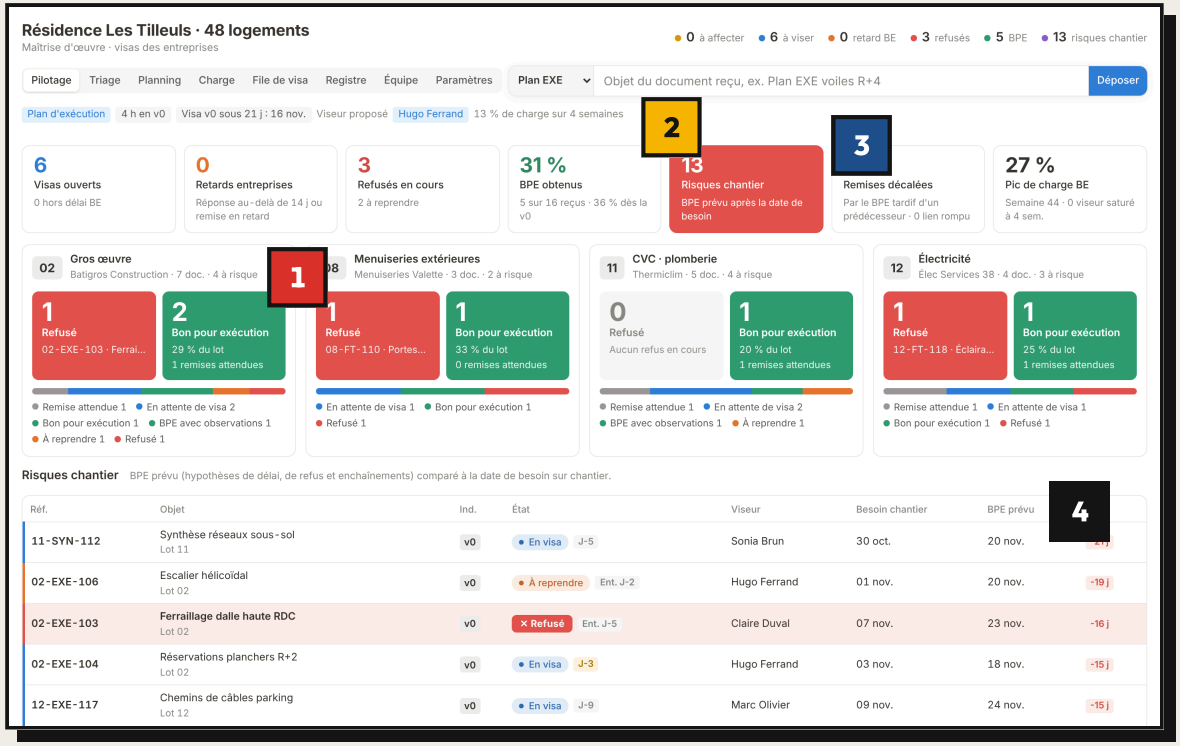
ENGINEERING FIRM · CONFIDENTIAL

1 Packages at a glance

Open rejections, approvals for construction and expected submissions, package by package and contractor by contractor.

2 Upload and assign

A received document is logged in one line. A reviewer is suggested based on discipline and workload over four weeks.



Overview screen. Fictitious project, contractors and people.

3 Site risks

The planned approval date is compared with the date the site needs it. What will hold up the works shows up early.

4 The gap in days

Each document shows its revision, reviewer, status and the gap between planned approval and site need.

WHAT I DESIGNED

An engineering firm's drawing approval register: document upload and assignment, review queue per reviewer, revision planning with rejection probability, workload curve, dependencies between documents, alert when an approval risks arriving after the site needs it.

TOOLS

Single-page web app, real-time shared database, approval lead-time forecasting.

Box ABCD

The competition planned in 3D, from floor plan to leaderboard

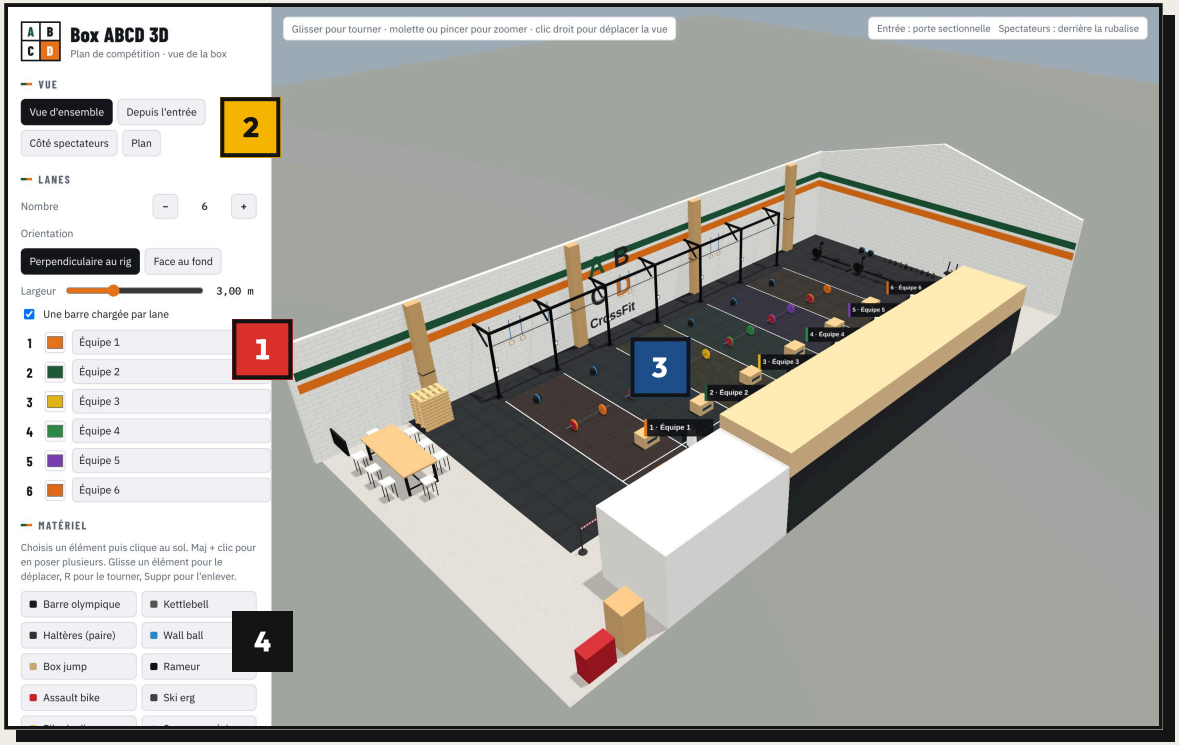
CROSSFIT GYM · COMPETITIONS

1 Lanes and teams

Number of lanes, width, orientation and each team's color: the floor redraws with every setting.

2 Viewpoints

Overview, from the entrance, from the spectator side or top-down: enough to brief judges and volunteers.



Floor setup view. Anonymized box and teams.

3 The box to scale

Rig, loaded barbells, box jumps and spectator area placed to scale, to settle the layout before the big day.

4 Equipment to place

Pick an item and click on the floor. Equipment planned, set up and checked is tracked heat by heat.

WHAT I DESIGNED

A competition tool for a CrossFit box: 3D floor to scale, lanes and teams, equipment to place and check, projected heats and timer, live leaderboard, TV mode for spectators.

TOOLS

Web app, Three.js for 3D, real-time shared database.